ARTIFICIAL INTELLIGENCE

EXPLORE · REASON · SEARCH · SOLVE

ARTIFICIAL INTELLIGENCE

MAKING MACHINES THINK



Simulation of human intelligence processes by machines.

- Learning from experience
- Reasoning and problem solving
- Perceiving the environment
- Making intelligent decisions

INFORMED VS UNINFORMED SEARCH

TWO WAYS TO EXPLORE A SEARCH SPACE

UNINFORMED SEARCH
(BLIND SEARCH)



- Uses no additional information about the goal.
- Explores nodes systematically (BFS, DFS, etc.).
- May explore many unnecessary nodes

INFORMED SEARCH
(HEURISTIC SEARCH)



- Uses heuristic information to guide the search.
- Explores more promising nodes first (A*, Greedy Best First, etc.).
- More efficient and goal-driven search.

LOCAL SEARCH

FINDING THE BEST NEARBY SOLUTION



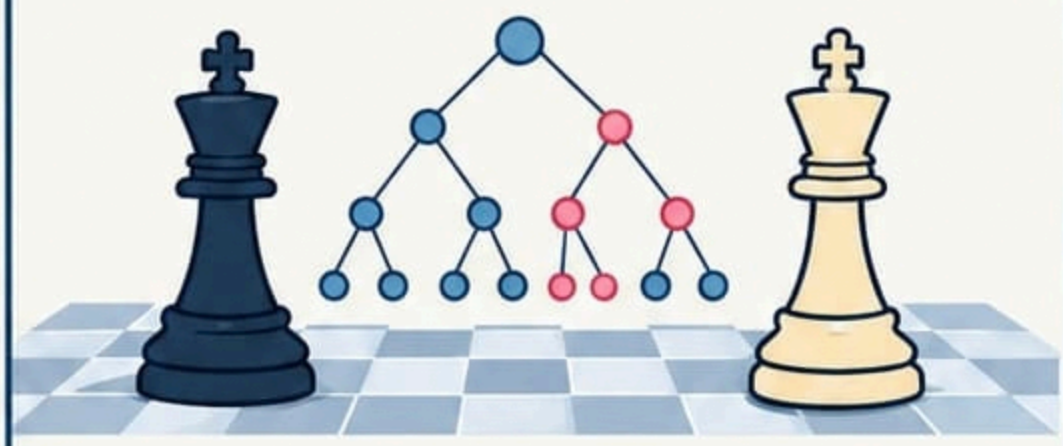
Local search algorithms operate on a single current state and move to a neighbouring state based on an evaluation function to find an optimal (or near-optimal) solution.

COMMON TECHNIQUES

- Hill Climbing
- Simulated Annealing
- Local Beam Search
- Genetic Algorithms

ADVERSARIAL SEARCH

DECISION MAKING IN A COMPETITIVE ENVIROMENT



Adversarial search is used in competitive environments with multiple intelligent agents having conflicting goals (e.g., games).

KEY CONCEPTS

- Game Tree
- Minimax Algorithm
- Alpha-Beta Pruning
- Optimal Strategy